BIN6001/IFT 6292 - PyMOL (20%)

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PDB: 6U79

5’ UTR stem loop B from West Nile Virus

The 5' untranslated region (UTR) of the West Nile Virus (WNV) contains conserved RNA structures essential for genome cyclization and replication. One such structure is stem-loop B (SLB), a short, variable hairpin with multiple key roles. Firstly, SLB contains the AUG initiation codon making it a critical site for translation initiation. Within the SLB structure is the 5'-upstream AUG region (5’-UAR). This element facilitates genome cyclization by pairing with a complementary 3’-UAR on the opposite end of the entire genome. Adjacent to SLB, the 5’-UAR-flanking stem (UFS) stabilizes the 5'-UAR/SLB region by forming a uracil-rich duplex which regulates RNA-dependent RNA polymerase (RdRP) recruitment. NS5 polymerase contains an RdRP region and recognizes SLB as a binding site. In the linear genome conformation, this duplex keeps SLB sequestered, preventing premature interactions. However, as the genome circularizes, the duplex dissociates, triggering SLB unfolding and enabling long-range base pairing between the 5'-UAR and 3'-UAR sequences. This is crucial for initiation of viral replication. Mutations affecting this region hinder structural transitions, compromising replication efficiency of the virus, making it a potential target for small-molecule antiviral therapies.

PDB: 6SX0

Wild-type H7N1 NS1 RNA-binding domain complexed with dsRNA

Non-structural protein 1 (NS1) is a viral protein of influenza A viruses that plays a key role in viral replication and immune evasion. The 6SX0 structure focuses on a specific RNA-binding domain of wild-type H7N1 NS1 complexed with the synthetic palindromic dsRNA AWFC01, which contains GUAAC motifs at each blunt end. The RNA-binding domain's structure enables it to bind with high affinity to virus-specific RNA motifs like GUAAC, allowing NS1 to modulate the structure and function of viral and host RNAs, affecting multiple stages of infection. By binding with viral mRNAs, such as those encoding M1 and NS1, it can regulate splicing and translation to enhance viral gene expression. Additionally, its affinity for dsRNA allows it to sequester host signaling molecules such as RIG-I from detecting viral RNA that would otherwise trigger antiviral defenses downstream.

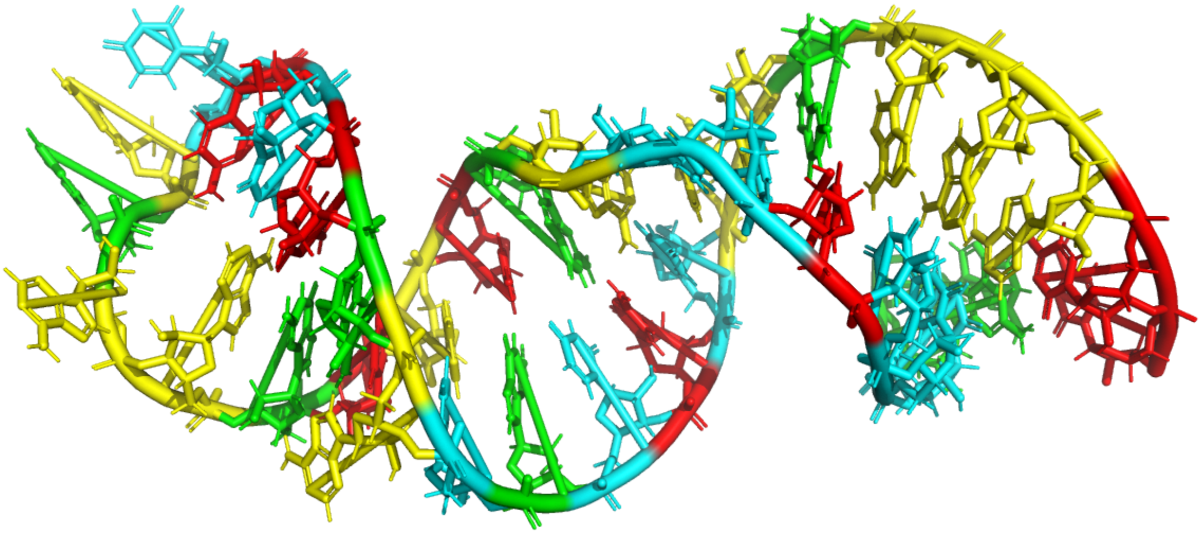

Image 1: 5' UTR stem loop B from West Nile Virus (RNA) in sticks model representation with coloured nucleotides - Guanine (green), Cytosine (red), Adenine (yellow), Uracil (cyan)

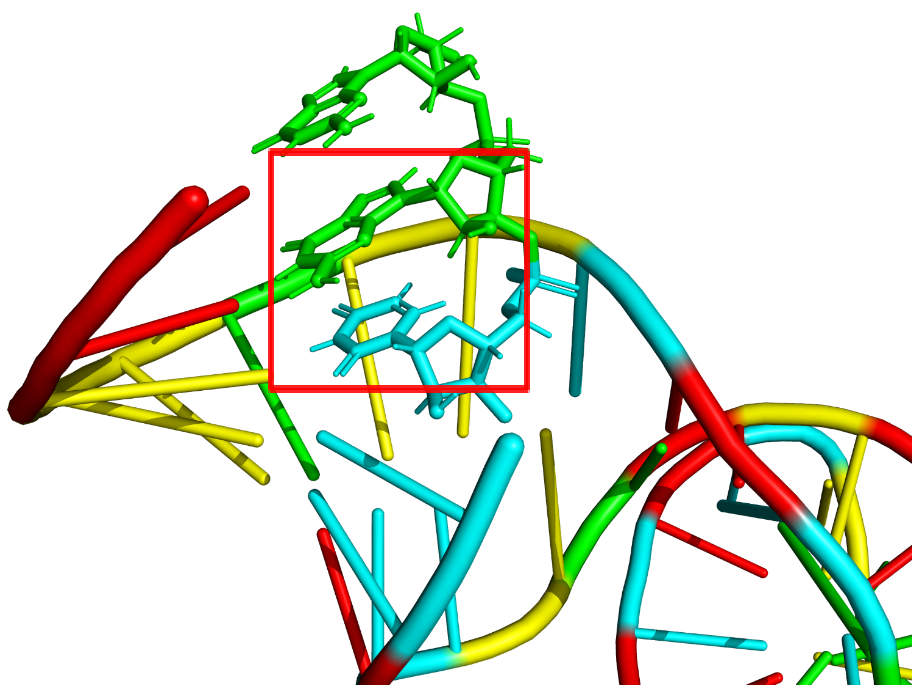

Image 2: Stacked base pairs Guanine (green) and Uracil (cyan) at positions 2 and 3 respectively of the 5' UTR stem loop B sequence from the West Nile Virus

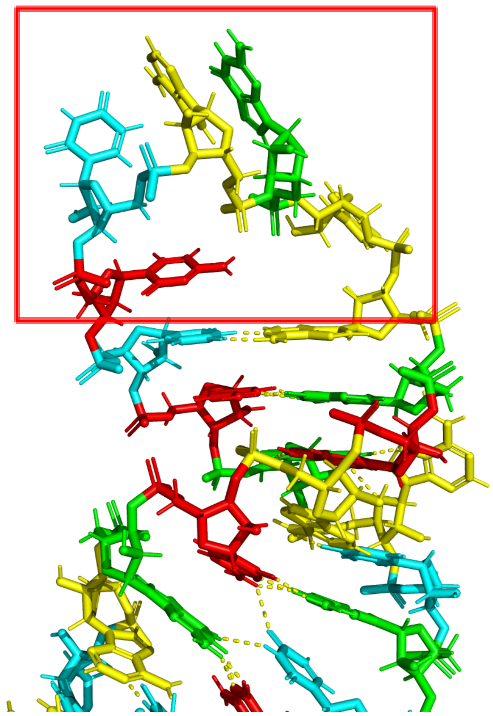

Image 3: Stem loop of the 5’ UTR stem loop B sequence from the West Nile Virus (left) with 5 unpaired nucleotides AGAUC in positions 16 - 20.

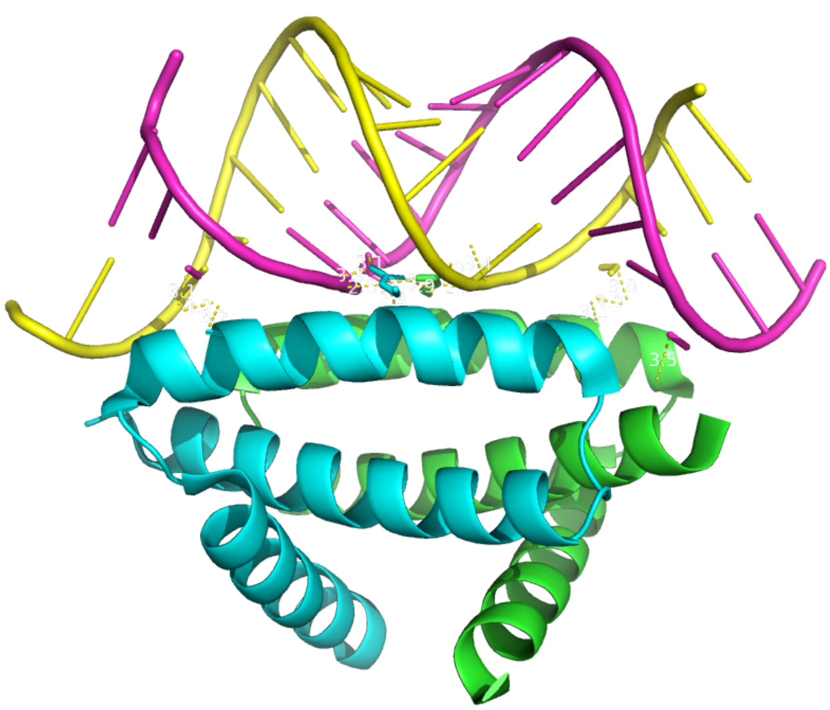

Image 4: Differently-coloured chains of the Wild-type H7N1 NS1 RNA-binding domain (Chain A: green, Chain B: cyan) complexed with dsRNA (Chain C: pink, Chain D: yellow).

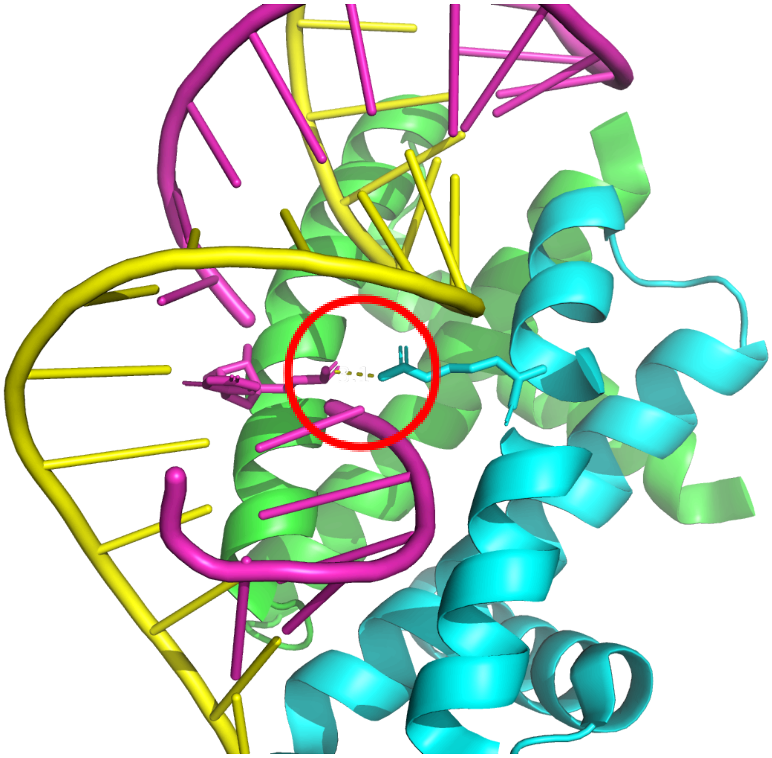

Image 5: Hydrogen bond (3.1 Å) between R38 of Chain B of the Wild-type H7N1 NS1 RNA-binding domain and U7 of the dsRNA strand.

<u>References</u>

De Falco, L., Silva, N. M., Santos, N. C., Huber, R. G., & Martins, I. C. (2021). The Pseudo-Circular Genomes of Flaviviruses: Structures, Mechanisms, and Functions of Circularization. *Cells*, *10*(3), 642. https://doi.org/10.3390/cells10030642

Marc, D., Barbachou, S., & Soubieux, D. (2013). The RNA-binding domain of influenzavirus non-structural protein-1 cooperatively binds to virus-specific RNA sequences in a structure-dependent manner. *Nucleic Acids Research*, *41*(1), 434–449. https://doi.org/10.1093/nar/gks979

Sharma, S., & Varani, G. (2020). NMR structure of Dengue West Nile viruses stem-loop B: A key cis-acting element for flavivirus replication. *Biochemical and Biophysical Research Communications*, *531*(4), 522–527. https://doi.org/10.1016/j.bbrc.2020.07.115

Wacquiez, A., Coste, F., Kut, E., Gaudon, V., Trapp, S., Castaing, B., & Marc, D. (2020). Structure and Sequence Determinants Governing the Interactions of RNAs with Influenza A Virus Non-Structural Protein NS1. *Viruses*, *12*(9), Article 9. https://doi.org/10.3390/v12090947